

Computer Systems

COMP1071: Durham University

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Part I

Digital Electronics

1 Circuits

1.1 Transistors

Definition 1.1.1: Transistors are electronically controlled switches that turn on or off when voltage or current is applied to a control terminal.

Definition 1.1.2: A nMOS transistor is a sandwich of layers of conducting and insulating material. The gate is a conducting layer, on top of an insulating layer, on top of the substrate which is a semi-conductor. When its gate g is at a high voltage, current is allowed to flow from source to drain, and this is prevented when the gate is at a low voltage.

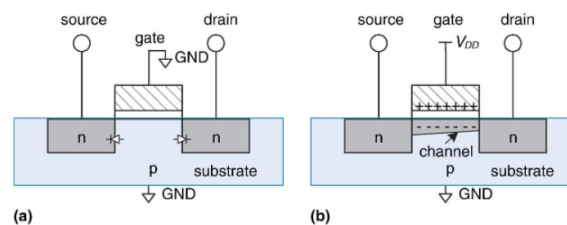


Figure 1: An nMOS transistor

The junction between p-type and n-type silicon is a diode. This means the current can only flow from p-type to n-type as they act as anodes and cathodes respectively.

Definition 1.1.3: A pMOS transistor is the opposite of a nMOS transistor - it allows current to flow when the gate is at low voltages, and prevents current flow at high voltages.

Definition 1.1.4: The static discipline states that if inputs to transistors meet valid thresholds, then the system must guarantee outputs will meet valid output thresholds and will not fall into the forbidden zone (outside of the accepted range for high/low voltages).

Definition 1.1.5: Moore's Law is usually quoted as "transistor density doubles every two years", or "computer processing power doubles every eighteen months".

1.2 Circuit Design

Definition 1.2.3: Abstraction is used when designing circuits to simplify problems by hiding unimportant details, highlighting the actual logic problem to be solved.

Definition 1.2.4: Another design principle used is discipline. Here, design choices are deliberately restricted to make circuits easier to model, design, and combine.

Definition 1.2.5: When designing complex systems, engineers use the three-y's: hierarchy, dividing a system into clear modules and submodules; modularity, using well-defined functions and interfaces for modules; and regularity, encouraging uniformity so modules can be swapped or reused.

1.3 Simple Circuits

2 Introduction to Logic

2.1 Functionally Complete Sets

Definition 2.0.1: A functionally complete set is a set of boolean operators which can be used to express all possible truth tables by combining members of the set into a boolean expression. For example, $\{AND, OR, NOT\}$, $\{NOR\}$, and $\{NAND\}$ are all functionally complete sets. This is demonstrated by

$AANDB$	$AORB$	$NOTA$
test	test	test

Table 1: Proving functional completeness

Part II

Machine Architecture

Part III

Databases

Part IV

Operating Systems

Part V

References

3 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Digital Electronics* and *Machine Architecture* are taught by Ioannis Ivrissimtzis.
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 - **Preferred Contact Methods:** ioannis.ivrissimtzis@durham.ac.uk
 - **Pronouns:** he, him
- *Databases* is taught by Peter Davies-Peck.
 - **Office Location:** Room 2063, MCS Building
 - **Preferred Contact Methods:** peter.w.davies@durham.ac.uk
 - **Pronouns:** he, him
- *Operating Systems* is taught by Anish Jindal.
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4 Cited Works